

Lloyd Institute of Engineering & Technology

Department of Computer Science & Engineering (AIML) | Session 2026-27

MINI PROJECT - WEEKLY PROGRESS LOG

Student Name	Dev Rana Kaila	Roll Number	2301531530023
Branch / Semester	B.Tech CSE (AIML) - Sem VII	Section	A
Project Guide	Dr. Devendra Gautam	Designation	Associate Professor, Dept. of CSE
Project Title	Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection		
GitHub	github.com/DevRanaKaila/DeepGuard-AI	College	LIET, Greater Noida

Weekly Activity Log (Guide Signatures Required)

Week	Date Range	Phase / Milestone	Activities & Deliverables	Status	Guide Signature
Week 1		Topic Finalisation & Literature Study	Finalised project topic on deepfake detection. Studied 8 research papers (Rossler, Frank, Qian, Afchar, Li, Tan, Selvaraju, Durall). Identified FaceForensics++ and Celeb-DF v2 as benchmark datasets. Set up Python environment: PyTorch, timm, OpenCV, Streamlit.	Completed	
Week 2		Synopsis Preparation	Prepared formal MINI PROJECT SYNOPSIS in LIET-prescribed format. Defined 6 project objectives. Drafted Literature Survey, Work Plan, Hardware/Software requirements. Submitted draft to Dr. Devendra Gautam for review.		
Week 3		Synopsis Approval & Architecture Design	Incorporated mentor feedback. Received synopsis approval. Designed dual-stream pipeline diagram, DFD Level-0 and Level-1. Created GitHub repository: github.com/DevRanaKaila/DeepGuard-AI. Initial commit with project structure.		
Week 4		Dataset Curation & Preprocessing Pipeline	Set up FaceForensics++ dataset structure. Implemented video frame extraction at 5 FPS using OpenCV. Integrated MTCNN for face detection with 15% boundary margin. Verified preprocessing pipeline on 50 sample videos. Created src/data/ module.		
Week 5		Spatial Stream Implementation	Integrated EfficientNet-B4 backbone via timm (pretrained ImageNet). Implemented SpatialStream module in src/models/spatial_stream.py. Validated 1792-d feature vector output and gradient flow. Spatial-only baseline achieves 82% accuracy.		
Week 6		DCT Transform & Mid-Term Review	Implemented 2D-DCT-II transform and Radial Azimuthal Spectrum A(r) in src/data/dct_transform.py. Demonstrated spatial pipeline to mentor. Prepared mid-term progress report. Frequency stream CNN prototype initiated.		

Week 7		Frequency Stream & Fusion Module	Implement 4-stage Spectral-CNN (src/models/frequency_stream.py). Implement attention-gated bilinear fusion (src/models/fusion.py). Complete DualStreamNet integration. Begin full training on FF++ c23 split.		
Week 8		Dashboard, Grad-CAM & Testing	Develop Streamlit web dashboard (app.py). Implement Grad-CAM heatmap generator (src/utis/gradcam.py). End-to-end pipeline integration. Cross-dataset testing on Celeb-DF v2. Benchmark against XceptionNet and F3-Net.		
Week 9		Report Writing & Code Finalization	Draft complete 60+ page project report. Conduct ablation study (spatial-only vs. dual-stream). Run plagiarism check (Turnitin <20%). Finalize GitHub repo with full documentation and README.		
Week 10		Final Submission & Viva Defense	Spiral-bind project report. Prepare 11-slide viva presentation. Live demo of Streamlit dashboard with real and fake test videos. Final submission to Dr. Devendra Gautam. Panel Q&A defense.		

Note: Weeks marked Completed have been signed by the Project Guide. Remaining weeks will be signed upon completion of respective activities.

Student Signature: _____

Guide Signature: _____

Dev Rana Kaila | Roll No. 2301531530023

Dr. Devendra Gautam | Assoc. Professor, CSE